MadLoop: Ninja and NinjaRS

Standard Model · reduction, total time and numerical stability

2026-10-22 · actual measured components

Three measurements answer distinct questions. The stock MG7 CLI reports an estimated reduction stage, a numerator-stage reference, and total time per phase-space point for the current exports. A separate six-process test compares warmed, full-helicity, rescue-enabled MadLoop against original revision `27147b3300f5`. Native-DP transformation tests compare Ninja and NinjaRS on identical random proposal streams with rescue disabled. Their samples, timing boundaries and error estimates are kept separate.

1 Methods and sample scopes

The main CLI table uses full helicity/color sums. Its **estimated reduction** is FULL–SKIP, including master integrals, callbacks and Laurent assembly. The **numerator stage (SKIP)** also contains non-reduction driver, Born and kinematic work; it is not an instruction-level tensor stopwatch. Adaptive traversal counts and raw floating timing metadata are retained for every stage. Three independent sessions per tool give empirical median-bootstrap intervals, with no small-sample coverage guarantee. All 72 sessions were retained. Ten processes have matching scopes across all six sessions. For the two on-shell SM cases, the selected timing helicity and filter changed after session 0; their per-tool medians remain shown, but the unchanged conservative reader suppresses aggregate ratios as NA. These direct-DP timings supply no timed rescue histogram. Optional polarized rows are shown only for positive saved helicity weight; a zero-weight selected configuration supplies no physical reduction-speed comparison.

The random-DP panels use 100 declared points for each process and tool. Born processes use full sums; $gg \to H$ and $gg \to HH$ use the preregistered all-minus helicity. On-shell $2 \to 1$ samples vary boosts at one physical invariant. Other cases use 1 TeV and a declared massive-RAMBO proposal with fixed cuts, rather than uniform massive phase-space integration. The empirical tail resolution is 1%; a zero count has an exact one-sided 95% binomial reference upper bound of 2.95%. A missing Ninja tail produces a ratio gap, without adding an epsilon.

Typography, margins, navy headings, pale panels and logarithmic green–white–red timing cells follow the NinjaRS report. Transformation comparisons follow Hirschi–Peraro the present finite sample is stated explicitly rather than identified with the paper's larger surveys.

2 MG7 CLI: estimated reduction and total time

2.1 SM: one or two final-state particles

Large values: **NinjaRS**. The comparison line gives Ninja; all times are in µs per point. Each interval uses the three actual sessions. Gray NA cells retain per-tool values without a cross-session ratio.

Process / actual calls	Estimated reduction	Numerator stage (SKIP)	Total
$u\bar{u} \to Z$ 3 sessions per tool Ninja N [10707, 3818, 4082]; NRS N [16376, 6549, 6443]	**16.9 µs** 95% CI 11.2–17.3 MAD 0.450 µs Ninja 35.0 µs **NA · scopes differ**	**14.5 µs** 95% CI 14.4–14.6 MAD 0.090 µs Ninja 14.5 µs **NA · scopes differ**	**31.2 µs** 95% CI 25.8–31.8 MAD 0.546 µs Ninja 49.5 µs **NA · scopes differ**
$u\bar{d} \to W^+$ 3 sessions per tool Ninja N [7315, 2629, 3186]; NRS N [10544, 4096, 4559]	**59.2 µs** 95% CI 56.8–70.6 MAD 2.40 µs Ninja 61.0 µs **NA · scopes differ**	**80.8 µs** 95% CI 80.6–82.0 MAD 0.198 µs Ninja 79.7 µs **NA · scopes differ**	**140 µs** 95% CI 137–153 MAD 2.60 µs Ninja 144 µs **NA · scopes differ**
$gg \to t\bar{t}$ 3 sessions per tool Ninja N [5581, 3361, 3378]; NRS N [6936, 4810, 4477]	**257 µs** 95% CI 224–267 MAD 10.5 µs Ninja 264 µs **×0.972** 95% CI 0.824–1.073	**879 µs** 95% CI 873–886 MAD 5.61 µs Ninja 878 µs **×1.001** 95% CI 0.982–1.016	**1135 µs** 95% CI 1110–1140 MAD 4.89 µs Ninja 1142 µs **×0.994** 95% CI 0.970–1.002
$e^+e^- \to \mu^+\mu^-$ 3 sessions per tool Ninja N [4889, 2544, 2641]; NRS N [5759, 3158, 3213]	**199 µs** 95% CI 192–209 MAD 6.77 µs Ninja 193 µs **×1.028** 95% CI 0.884–1.118	**533 µs** 95% CI 531–535 MAD 1.89 µs Ninja 534 µs **×0.997** 95% CI 0.972–1.003	**734 µs** 95% CI 723–742 MAD 8.17 µs Ninja 740 µs **×0.992** 95% CI 0.964–1.029

N lists the actual unpolarized traversals in sessions 0/1/2. FULL and SKIP use the same N; polarized stages use 2N. Exact per-session phase counts, selected helicities, filter hashes and raw timings remain in JSON.

2.2 SM: three or four final-state particles

Large values: **NinjaRS**. The comparison line gives Ninja; all times are in µs per point. Each interval uses the three actual sessions. Gray NA cells retain per-tool values without a cross-session ratio.

Process / actual calls	Estimated reduction	Numerator stage (SKIP)	Total
$u\bar{u} \to Zgg$ 3 sessions per tool Ninja N [565, 623, 618]; NRS N [658, 639, 663]	**2866 µs** 95% CI 2798–3058 MAD 67.5 µs Ninja 3735 µs **×0.767** 95% CI 0.455–1.153	**39 019 µs** 95% CI 35 239–39 206 MAD 187 µs Ninja 39 493 µs **×0.988** 95% CI 0.887–1.121	**42 004 µs** 95% CI 38 105–42 077 MAD 73.1 µs Ninja 42 373 µs **×0.991** 95% CI 0.881–1.023
$e^+e^- \to t\bar{t}\gamma$ 3 sessions per tool Ninja N [611, 654, 658]; NRS N [429, 699, 562]	**2787 µs** 95% CI 45.1–4963 MAD 2176 µs Ninja 2003 µs **×1.391** 95% CI 0.007–3.279	**31 711 µs** 95% CI 30 741–33 244 MAD 970 µs Ninja 32 326 µs **×0.981** 95% CI 0.919–1.034	**34 498 µs** 95% CI 33 289–35 704 MAD 1206 µs Ninja 34 969 µs **×0.987** 95% CI 0.862–1.045
$u\bar{u} \to ZZgg$ 3 sessions per tool Ninja N [50, 50, 50]; NRS N [50, 50, 50]	**52 200 µs** 95% CI 50 284–65 772 MAD 1916 µs Ninja 60 235 µs **×0.867** 95% CI 0.686–1.902	**1 052 194 µs** 95% CI 1 046 490–1 057 390 MAD 5196 µs Ninja 1 066 664 µs **×0.986** 95% CI 0.981–1.015	**1 109 590 µs** 95% CI 1 102 477–1 112 261 MAD 2672 µs Ninja 1 102 205 µs **×1.007** 95% CI 0.967–1.010
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ 3 sessions per tool Ninja N [50, 82, 81]; NRS N [50, 80, 83]	**29 547 µs** 95% CI 26 573–29 732 MAD 186 µs Ninja 24 369 µs **×1.212** 95% CI 0.825–1.354	**309 583 µs** 95% CI 307 000–311 387 MAD 1804 µs Ninja 310 122 µs **×0.998** 95% CI 0.989–1.006	**336 732 µs** 95% CI 336 156–340 934 MAD 576 µs Ninja 334 491 µs **×1.007** 95% CI 0.983–1.025

N lists the actual unpolarized traversals in sessions 0/1/2. FULL and SKIP use the same N; polarized stages use 2N. Exact per-session phase counts, selected helicities, filter hashes and raw timings remain in JSON.

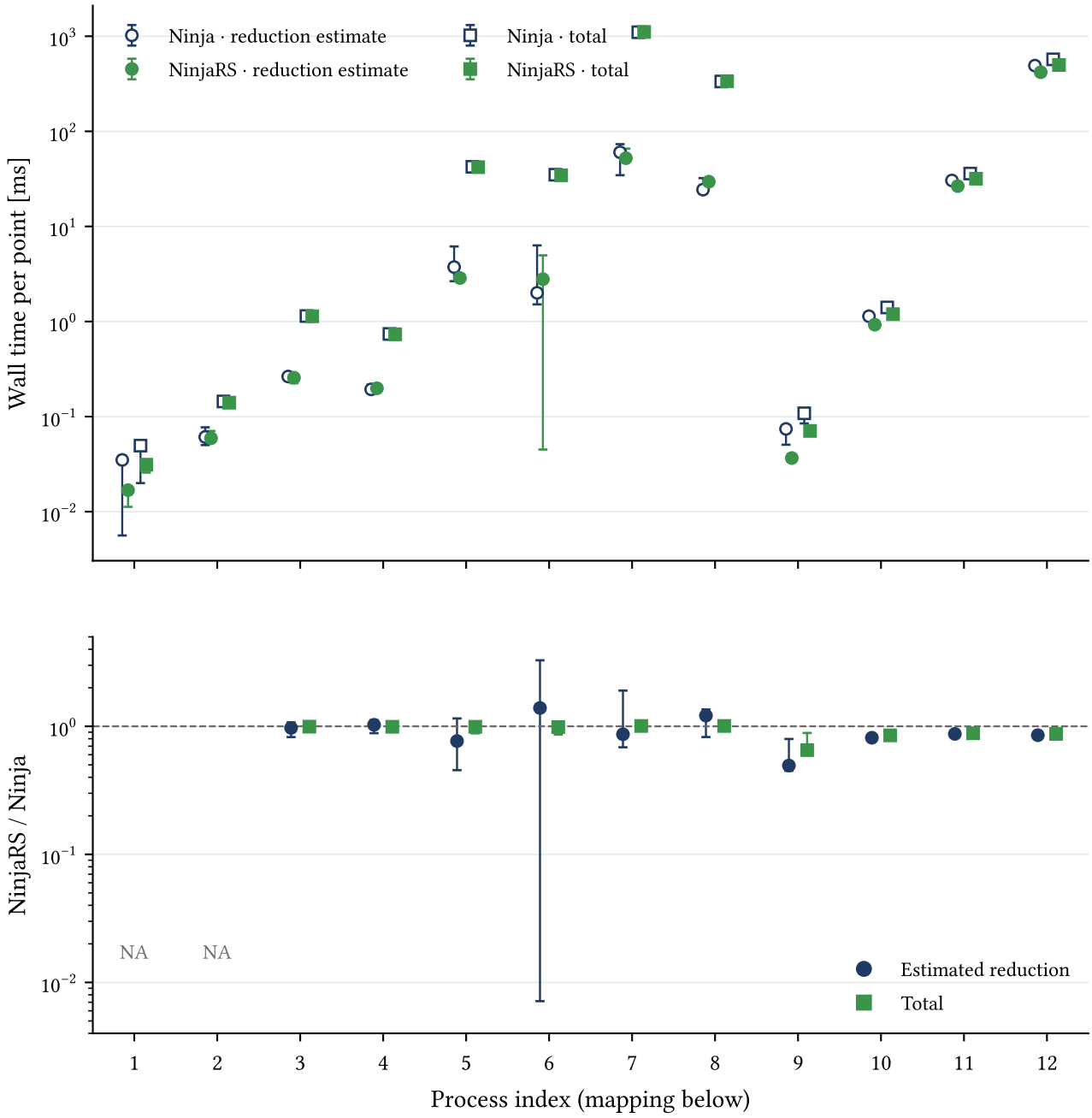
2.3 Loop-induced amplitudes: H, HH, HHH and HHHH

Large values: **NinjaRS**. The comparison line gives Ninja; all times are in µs per point. Each interval uses the three actual sessions. Gray NA cells retain per-tool values without a cross-session ratio.

Process / actual calls	Estimated reduction	Numerator stage (SKIP)	Total
$gg \to H$ 3 sessions per tool Ninja N [9984, 3790, 3900]; NRS N [14105, 6084, 5924]	**36.6 µs** 95% CI 33.9–40.3 MAD 2.71 µs Ninja 74.2 µs **×0.494** 95% CI 0.449–0.797	**34.7 µs** 95% CI 34.1–34.9 MAD 0.193 µs Ninja 34.1 µs **×1.017** 95% CI 0.995–1.023	**70.7 µs** 95% CI 68.6–75.2 MAD 2.09 µs Ninja 108 µs **×0.652** 95% CI 0.626–0.888
$gg \to HH$ 3 sessions per tool Ninja N [5416, 3172, 3171]; NRS N [7404, 4797, 4616]	**926 µs** 95% CI 919–939 MAD 6.37 µs Ninja 1137 µs **×0.814** 95% CI 0.763–0.828	**269 µs** 95% CI 269–270 MAD 0.210 µs Ninja 269 µs **×1.001** 95% CI 0.995–1.007	**1195 µs** 95% CI 1189–1208 MAD 5.53 µs Ninja 1406 µs **×0.850** 95% CI 0.807–0.860
$gg \to HHH$ 3 sessions per tool Ninja N [703, 663, 664]; NRS N [804, 807, 813]	**26 557 µs** 95% CI 25 824–26 909 MAD 353 µs Ninja 30 429 µs **×0.873** 95% CI 0.842–0.892	**5585 µs** 95% CI 5199–5697 MAD 112 µs Ninja 5787 µs **×0.965** 95% CI 0.895–1.028	**31 755 µs** 95% CI 31 408–32 606 MAD 347 µs Ninja 35 972 µs **×0.883** 95% CI 0.861–0.907
$gg \to HHHH$ 3 sessions per tool Ninja N [50, 50, 50]; NRS N [53, 54, 54]	**418 630 µs** 95% CI 418 480–418 796 MAD 150 µs Ninja 491 052 µs **×0.853** 95% CI 0.849–0.853	**79 966 µs** 95% CI 79 140–80 711 MAD 745 µs Ninja 79 617 µs **×1.004** 95% CI 0.983–1.027	**498 762 µs** 95% CI 497 770–499 191 MAD 429 µs Ninja 571 494 µs **×0.873** 95% CI 0.871–0.875

N lists the actual unpolarized traversals in sessions 0/1/2. FULL and SKIP use the same N; polarized stages use 2N. Exact per-session phase counts, selected helicities, filter hashes and raw timings remain in JSON.

3 MG7 profile: process-indexed timings



Circles denote the estimated reduction stage and squares the total. Open markers are Ninja; filled markers are NinjaRS. Bars give empirical independent-session 95% intervals. No line connects unrelated process families. The ratio panel preserves the large uncertainty of some three-session SM estimates; QCD1/EW1 remain NA because the selected helicity/filter changed between sessions. The numerator stage and exact phase counts are given in the preceding tables.

1	$u\bar{u} \to Z$	2	$u\bar{d} \to W^+$
3	$gg \to t\bar{t}$	4	$e^+e^- \to \mu^+\mu^-$
5	$u\bar{u} \to Zgg$	6	$e^+e^- \to t\bar{t}\gamma$
7	$u\bar{u} \to ZZgg$	8	$e^+e^- \to \mu^+\mu^-\gamma\gamma$
9	$gg \to H$	10	$gg \to HH$
11	$gg \to HHH$	12	$gg \to HHHH$

4 Original MadLoop comparison: six warmed processes

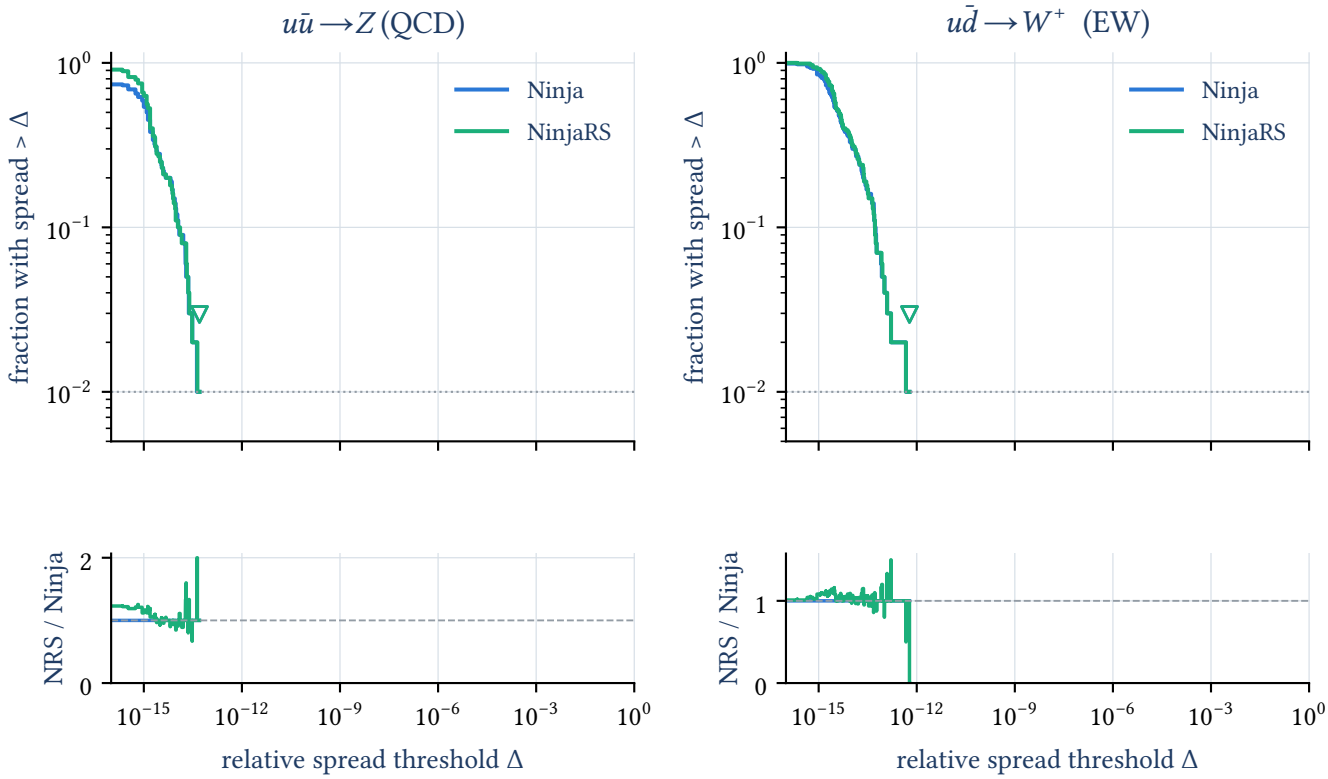
Each row uses the same ten fixed prefix points, seven adjacent alternating pairs, full helicity/color sums, poles and normal rescue. The denominator is original MadLoop with Ninja at `27147b3300f5` the numerator is the installed NinjaRS cohort in the provenance records. A 1.05 ratio is shown as a reference for this sample. The former heavy eight-process runtime requirement was withdrawn when the measurement scope was narrowed.

Process	Original Ninja µs / point	NinjaRS µs / point	NinjaRS / original Ninja
$u\bar{u} \to Z$ 10 points · 7 pairs	**42.1** MAD 0.071	**38.7** MAD 0.053	**×0.920** 95% CI 0.912–0.922
$u\bar{d} \to W^+$ 10 points · 7 pairs	**203** MAD 0.164	**190** MAD 0.145	**×0.934** 95% CI 0.930–0.941
$gg \to t\bar{t}$ 10 points · 7 pairs	**1385** MAD 3.24	**1223** MAD 1.86	**×0.883** 95% CI 0.880–0.887
$e^+e^- \to \mu^+\mu^-$ 10 points · 7 pairs	**799** MAD 2.10	**826** MAD 1.75	**×1.033** 95% CI 1.031–1.039
$gg \to H$ 10 points · 7 pairs	**158** MAD 2.78	**124** MAD 0.175	**×0.786** 95% CI 0.784–0.799
$gg \to HH$ 10 points · 7 pairs	**2804** MAD 6.27	**2263** MAD 13.5	**×0.807** 95% CI 0.790–0.815

The 0.5 s setting is the calibration target used once to select the loop count. Actual warmed commands lasted 0.326–0.519 s. Saved status vectors describe the final traversal of each command; they do not form a complete histogram over every adaptive-loop evaluation. All selected final traversals returned normal DP status for both tools.

5 Native-DP stability: On-shell 2→1

Unrecovered native double precision · direction + Lorentz transformation



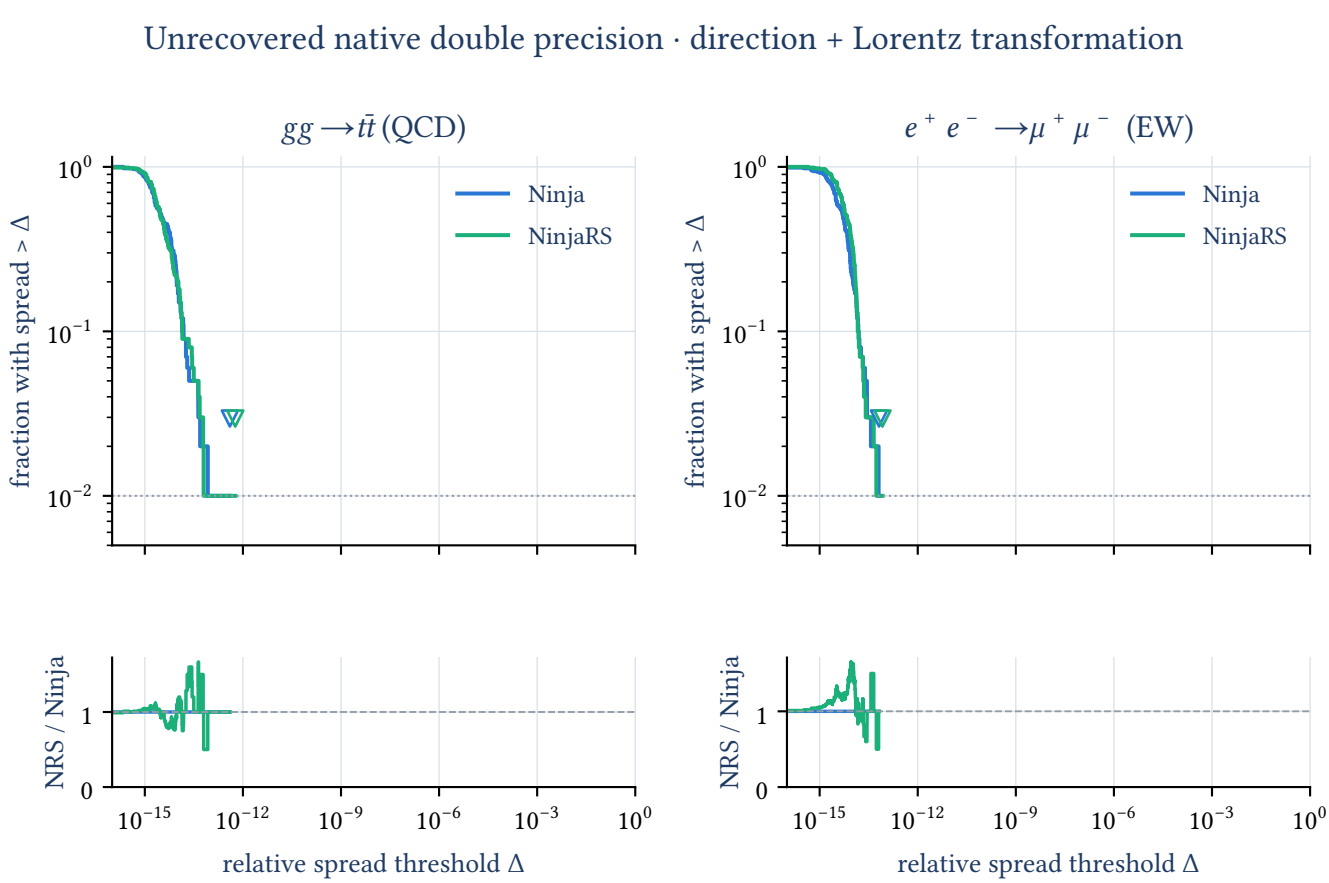
on-shell boosted frames; one physical invariant; Ecm=91.188 GeV; seed=20261021
no final-state cuts
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

on-shell boosted frames; one physical invariant; Ecm=80.419 GeV; seed=20261022
no final-state cuts
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

All N=100 observations; resolution 1%. Hollow triangles: zero events, one-sided 95% upper reference 2.95%.
Ninja-zero tail ratios are undefined/unbounded gaps; no epsilon or smoothing.
Self-consistency estimates do not measure error against a finite-part oracle.

The combined spread uses the same original, reversed-direction and Lorentz-transformed point. Hollow triangles mark zero-event upper references, not observed failures. All observations in this selected sample were finite. Full direction-only and Lorentz-only vector panels and exact tail counts accompany the report.

6 Native-DP stability: Two final-state particles



massive RAMBO proposal, not uniform massive phase space; Ecm=1000 GeV; seed=20261023
cuts: pT > 50 GeV; ΔR > 0.5
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

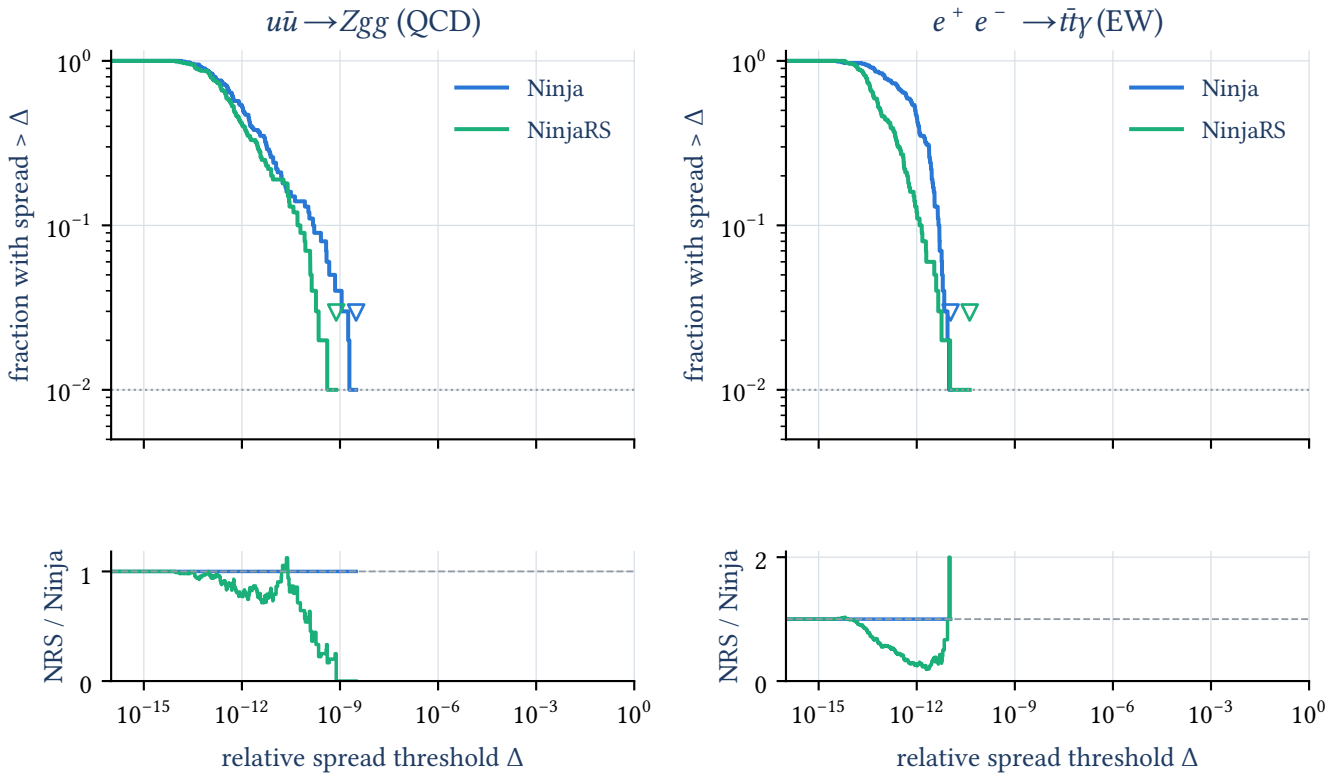
massive RAMBO proposal, not uniform massive phase space; Ecm=1000 GeV; seed=20261024
cuts: pT > 50 GeV; ΔR > 0.5
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

All N=100 observations; resolution 1%. Hollow triangles: zero events, one-sided 95% upper reference 2.95%.
Ninja-zero tail ratios are undefined/unbounded gaps; no epsilon or smoothing.
Self-consistency estimates do not measure error against a finite-part oracle.

The combined spread uses the same original, reversed-direction and Lorentz-transformed point. Hollow triangles mark zero-event upper references, not observed failures. All observations in this selected sample were finite. Full direction-only and Lorentz-only vector panels and exact tail counts accompany the report.

7 Native-DP stability: Three final-state particles

Unrecovered native double precision · direction + Lorentz transformation



massive RAMBO proposal, not uniform massive phase space; Ecm=1000 GeV; seed=20261025
cuts: pT > 50 GeV; ΔR > 0.5
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

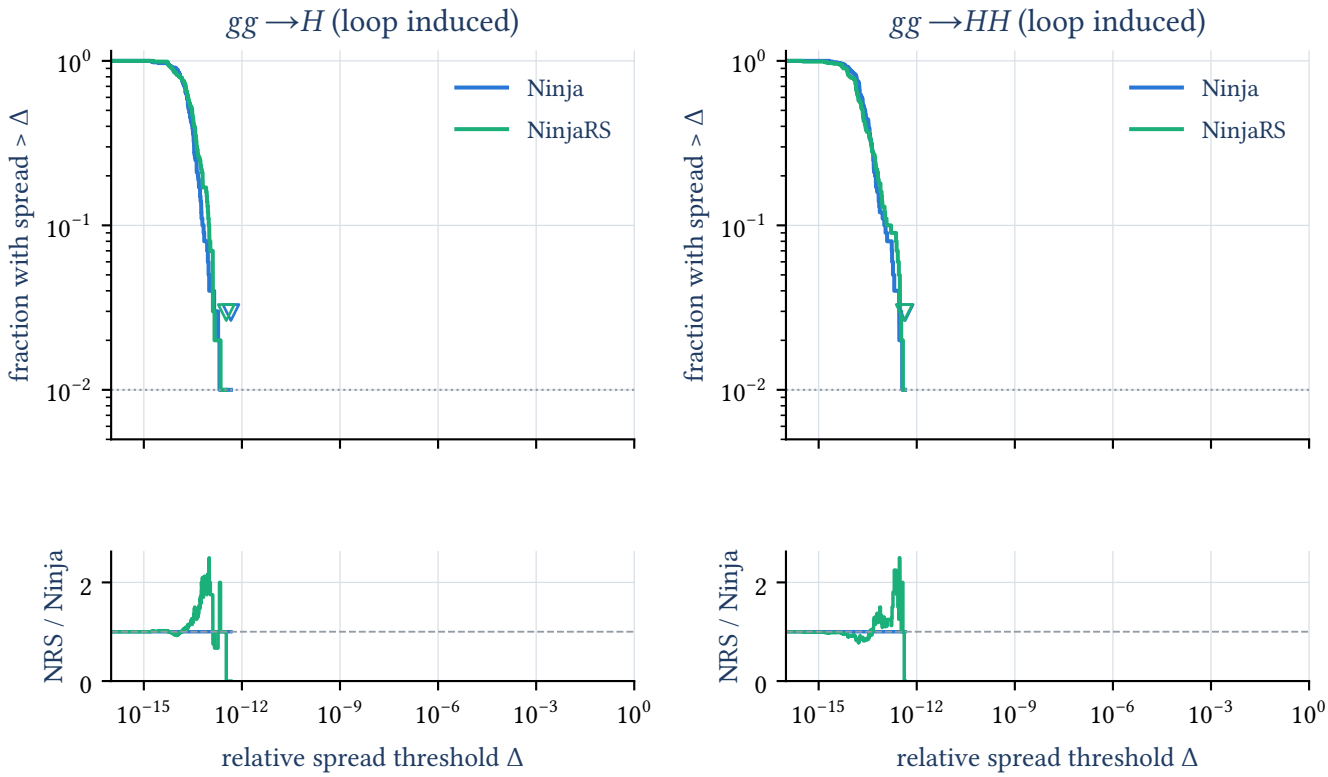
massive RAMBO proposal, not uniform massive phase space; Ecm=1000 GeV; seed=20261026
cuts: pT > 50 GeV; ΔR > 0.5
full helicity/color sum; selector -1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

All N=100 observations; resolution 1%. Hollow triangles: zero events, one-sided 95% upper reference 2.95%.
Ninja-zero tail ratios are undefined/unbounded gaps; no epsilon or smoothing.
Self-consistency estimates do not measure error against a finite-part oracle.

The combined spread uses the same original, reversed-direction and Lorentz-transformed point. Hollow triangles mark zero-event upper references, not observed failures. All observations in this selected sample were finite. Full direction-only and Lorentz-only vector panels and exact tail counts accompany the report.

8 Native-DP stability: Loop-induced amplitudes

Unrecovered native double precision · direction + Lorentz transformation



on-shell boosted frames; one physical invariant; Ecm=125 GeV; seed=20261029
no final-state cuts
single all-minus helicity; tuple (-1, -1, 0); index 1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

massive RAMBO proposal, not uniform massive phase space; Ecm=1000 GeV; seed=20261030
cuts: ΔR > 0.5
single all-minus helicity; tuple (-1, -1, 0, 0); index 1
Ninja: N=100; 0 failed/unbounded
NinjaRS: N=100; 0 failed/unbounded

All N=100 observations; resolution 1%. Hollow triangles: zero events, one-sided 95% upper reference 2.95%.
Ninja-zero tail ratios are undefined/unbounded gaps; no epsilon or smoothing.
Self-consistency estimates do not measure error against a finite-part oracle.

The combined spread uses the same original, reversed-direction and Lorentz-transformed point. Hollow triangles mark zero-event upper references, not observed failures. All observations in this selected sample were finite. Full direction-only and Lorentz-only vector panels and exact tail counts accompany the report.

9 Numerical observations and practical limits

Process	Finite / N Ninja · NRS	Largest spread Ninja · NRS	Saved invalid flags Ninja · NRS
$u\bar{u} \to Z$	100 / 100 100 / 100	5.02e-14 5.06e-14	0 · 0 300 calls / tool
$u\bar{d} \to W^+$	100 / 100 100 / 100	6.01e-13 5.99e-13	0 · 0 300 calls / tool
$gg \to t\bar{t}$	100 / 100 100 / 100	3.99e-13 5.82e-13	0 · 300 300 calls / tool
$e^+e^- \to \mu^+\mu^-$	100 / 100 100 / 100	6.55e-14 8.33e-14	0 · 300 300 calls / tool
$u\bar{u} \to Zgg$	100 / 100 100 / 100	3.11e-09 7.66e-10	0 · 300 300 calls / tool
$e^+e^- \to t\bar{t}\gamma$	100 / 100 100 / 100	1.08e-11 4.16e-11	0 · 300 300 calls / tool
$gg \to H$	100 / 100 100 / 100	4.61e-13 3.33e-13	0 · 0 300 calls / tool
$gg \to HH$	100 / 100 100 / 100	4.41e-13 4.14e-13	0 · 300 300 calls / tool

Saved hardware IEEE-invalid flags occur on 300 transformed observations for NinjaRS in $gg \to t\bar{t}$, $e^+e^- \to \mu^+\mu^-$, $u\bar{u} \to Zgg$, $e^+e^- \to t\bar{t}\gamma$ and $gg \to HH$. Plain-prefix checks reproduce the flags. Saved divide-by-zero/overflow, algorithmic FPE/rejections, nonfinite results and MP/compensated use are zero. Hardware flags are retained separately from numerical rejection by the frozen observer.

Transformation self-consistency and reference agreement do not provide a scalar finite-part analytic oracle. Analytic pole controls, higher-precision convergence, installation/export checks and software regressions retain their own concrete evidence and scope.

10 Numerical agreement and normal rescue

The core comparison includes both $2 \to 4$ processes. Born, finite, single-pole, double-pole and split-order coefficients passed the retained common-point tolerance. Every case has one fixed 100-point stream, calibration and two repeated observations; the coefficient rows and status histograms repeat exactly. The counts below describe one traversal of each unique stream, rather than treating repeated traversals as additional independent points.

SM process	Points	NinjaRS DP	NinjaRS QP rescue	Agreement
$u\bar{u} \to Z$	100	100	0	passed
$u\bar{d} \to W^+$	100	100	0	passed
$gg \to t\bar{t}$	100	100	0	passed
$e^+e^- \to \mu^+\mu^-$	100	100	0	passed
$u\bar{u} \to Zgg$	100	98	2	passed
$e^+e^- \to t\bar{t}\gamma$	100	98	2	passed
$u\bar{u} \to ZZgg$	100	76	24	passed
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	100	96	4	passed

Across these 800 unique points NinjaRS returned DP code 218 for 768 points and genuine automatic QP-rescue code 322 for 32. This normal-controller sample is separate from the rescue-disabled DP figures and from the CLI cache; its rescue frequencies are not inserted into timing cells that did not save full timed-call status histories.

The additional $gg \to HHH$ and $gg \to HHHH$ controls passed matched Ninja6/NinjaRS8 full-sum, full-pole, normal-rescue comparisons at two fixed physical points per process, with zero Born terms. Their four native calls succeeded. A versioned, read-only finalizer corrected an analysis-only Python import error after collection; it repeated no native call and preserved the original parent exit 1. This is N=2 correctness evidence, distinct from the requested CLI timing sessions and from random stability.

11 Integration, precision and software validation

NinjaRS is selected with `MLReductionLib=8` through process-prefixed Fortran wrappers and a versioned Rust C ABI. Installation uses `install ninjars` after updating the checkout, `install ninjars --update` installs its pinned sources and lockfile transactionally. The measured numerical source is `856e76e66961`, with NinjaRS `aa6a6307` and oneloopmaster `8ec256fc` the final CLI observer is `464b441fe04d`. The later NinjaRS main `ca1009e` contains report-only changes: all 582 non-report source entries match the measured pin. The final watch is dated 22 October 2026.

Installation / export	Actual CLI update and compiled installed smoke; relocated standalone DP/QP/normal-controller returns 110/120/218. FKS/NLO passed 20/20 pole checks at its documented 10^{-5} smoke tolerance, plus FKS-export soft/collinear checks with zero failures. This is separate from the stricter coefficient comparisons; no shower validation is claimed.
Focused physics	45 persistent/fresh comparison pairs, 5,400 Born/Laurent/split components, cover complex masses, mixed orders and near-singular automatic rescue. Nine additional fixed two-point controls cover explicit QP direction and reused/finalized state changes; the separate normal-HH anchor supplies the normal-controller comparison.
Genuine precision	The numerator and improved kinematics are recomputed in Fortran binary128. Forty-digit input records recover each original 113-bit value before exact widening; reduction and scalar masters use 256-bit arithmetic. Actual installed-ABI controls contain 64 calls. All 1,392 retained reduction requests converge at 256/384 bits, including complete Laurent, CC and same-basis R1 components; the largest scaled Laurent difference is 2.73×10^{-62}.
Software	The integrated selected-source campaign ran 293 tests: 292 passed and one unchanged conditional skip requiring a generated QCD2 tree. The final observer/resource-path successor passed 95 affected software checks. These overlapping software scopes are not summed as independent tests, and do not replace the physical campaigns.

The standard Rust suite passed 24 checks with 7 ignored. When explicitly enabled, two ignored diagnostics retain the historical bit-identity mismatch and the unresolved exceptional direct-provider case.

The 256-bit calculation does not make the Fortran numerator 256-bit accurate. Returning through 40-digit output records can round to an adjacent binary128 value: 16 of 22,272 examined fields showed this, with largest Laurent-scaled difference 4.20×10^{-36}. Historical replay is deliberately not labelled bitwise identical: 468 binary128 records changed, and the original bit-identity test exit 101 remains retained. The maximum scaled complete-Laurent difference from historical outputs is 1.48×10^{-34}; a much larger basis-dependent R1 diagnostic change is not a change of that physical Laurent coefficient.

Direct scalar-provider exceptional-input controls still retain two unsafe successes upstream. The tested full-bridge planar calls were accurate in DP or safely rejected in MP; that bridge behavior does not resolve the direct-provider finding. The retained source-bound receipts distinguish this limitation from accepted MadLoop amplitudes and from the finite-sample stability plots.

12 Measured Fortran work and retained research

The accepted allocation change measured a 3.15% whole-process improvement in the historical QCD3 study (ratio 0.9685, 95% interval 0.9569–0.9936). That study used generated source `f4bffc61` and archive pair `832fc0d1` → `366ae91d`, distinct from the final installed `b0d69deb` archive. Its source-body attribution and exact records accompany the report. Earlier 9.24% DP callback-kernel and 3.19% Ninja6 Fortran-QP prototype results have different boundaries and are not additive. A separate 1,392-request Float256 native/interpreted screen had intervals crossing parity and was not promoted. The actual final timings above, including the slower EW2 ancillary ratio, determine the reported performance rather than these historical experiments.

13 First-session output, compilation and initialization

The following captured CLI costs provide preparation context for the per-point timings. Every cell gives **Ninja6 above NinjaRS8**, in seconds, for session 0 of that tool and process. These are direct metadata values, rounded to three significant digits for display; exact values and source hashes are in `final-report-v1/preparation-costs.csv`. They are not three-session statistics.

Process	Output / export	HELAS / model compile	Initial process compile	Initialization	Boot
$u\bar{u} \to Z$	0.849 0.795	2.65 2.56	6.2 6.37	0.0224 0.0147	0.00956 0.00927
$u\bar{d} \to W^+$	1.63 1.6	16.7 17.3	7.05 6.22	0.0328 0.0228	0.0176 0.0152
$gg \to t\bar{t}$	2.04 1.52	4.25 4.56	13.9 14	0.043 0.0346	0.00978 0.0101
$e^+e^- \to \mu^+\mu^-$	3.52 3.43	36 35.9	7.61 7.37	0.0491 0.0417	0.0189 0.0184
$u\bar{u} \to Zgg$	2.88 2.47	7.89 8.59	16.8 16.3	0.425 0.365	0.0108 0.0103
$e^+e^- \to t\bar{t}\gamma$	6.79 6.63	79.1 76.8	15.7 16.3	0.393 0.561	0.0269 0.0291
$u\bar{u} \to ZZgg$	6.75 6.77	8.85 9.31	24.6 24.1	14.3 13.8	0.0229 0.0247
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	20 20.4	42.4 43.7	31.5 31.4	6.31 7.3	0.0729 0.0738
$gg \to H$	0.714 0.864	2.05 2.2	10.9 11.4	0.024 0.017	0.00989 0.00911
$gg \to HH$	0.86 1.05	2.17 2.35	13.8 13.7	0.0443 0.0324	0.01 0.0105
$gg \to HHH$	1.36 2.14	2.83 2.78	19 17	0.342 0.299	0.0125 0.0129
$gg \to HHHH$	5.35 5.53	2.31 2.47	22.7 23.6	5.22 4.59	0.179 0.181

The exact captured keys are `Process_output`, `HELAS_MODEL_compilation`, `Process_compilation`, `Initialization` and `Booting_time`, respectively. Output/export is not labelled as standalone symbolic generation. The costs have different boundaries and are not asserted to form an additive total. Additional timed-stage recompilations and complete phase metadata remain in the raw records; boot subtraction and adaptive traversal counts retain their original CLI meanings.

14 Provenance and reproduction

The final profile study contains 72 accepted command records: twelve processes, three sessions and tools 6/8. The first session creates a 20-point CLI DP cache, with zero QP entries; later sessions reuse it. This cache is not a matched-point random-stream authority or a complete timed-call status history. Three earlier Ninja-only pilot attempts failed before a complete profile result; their distinct source/specification, error and cleanup records are retained separately and their partial timing fragments are excluded.

The full-sum CLI comparison includes both SM $2 \to 4$ processes and loop-induced H, HH, HHH and HHHH. The separate normal-rescue six-process comparison uses original Ninja revision `27147b3300f5` and final NinjaRS runtime `a7577a9a3329` / archive `b0d69deb8c16`. It passes the agreed 1.05 upper-interval reference in that amended ten-point, seven-pair scope. No canceled heavy eight-process timing requirement is reinstated, and no uniform SM speedup is inferred from the mixed, uncertain CLI estimates.

Offline PDF reproduction: from the repository root, run `python -B doc/benchmarkfolder/final-report-v1/reproduce.py --output reproduced.pdf`. It verifies the document's finite data/vector/font/style closure and compiles with Typst 0.15.1, bundled Libertinus fonts, a fixed SOURCE_DATE_EPOCH and creation timestamp. No native archive, physics compilation, network or amplitude evaluation is needed.

The source and raw custody verification is a separate operation requiring the exact retained installation and the concrete authority readers. Regenerating from pinned sources with a different compiler can produce new executable and measurement identities. The reproduction README links the immutable scientific archive at 19d67c860533, including the exact CLI cards, scalar coefficient records, timing metadata and prior partial report. This folder retains the finite offline document closure.

Styling is attributed to NinjaRS ca1009e and the earlier pyamplicol report. Tail and timing comparisons are inspired by Hirschi–Peraro, arXiv:1604.01363. The external optional plotting archive was inspected for methodology; no external plotting code is included. Libertinus 7.051 is distributed unmodified under the bundled SIL Open Font License.